

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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- System functions of discrete LTI systems
- System functions and unit impulse response
- System Functions and System Stability
- System functions and system frequency response

➢Definition of System Functions

If the difference equation describing a discrete LTI system is

$$\sum_{n=0}^{N} a_n y[k-n] = \sum_{n=0}^{M} b_n x[k-n]$$

By utilizing the displacement characteristics of the *z*-transform, the complex frequency domain equation describing the system can be obtained

$$(\sum_{n=0}^{N} a_n z^{-n}) Y(z) = (\sum_{n=0}^{M} b_n z^{-n}) X(z)$$

➢Definition of System Functions

$$(\sum_{n=0}^{N} a_n z^{-n}) Y(z) = (\sum_{n=0}^{M} b_n z^{-n}) X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{n=0}^{M} b_n z^{-n}}{\sum_{n=0}^{N} a_n z^{-n}}$$

➢The representation form of system functions

(1) The rational polynomial representation of z^{-1}

$$H(z)=\frac{b_0+b_1z^{-1}+\cdots+b_Mz^{-M}}{a_0+a_1z^{-1}+\cdots+a_Nz^{-N}}$$

(2) Rational polynomial representation of z

$$H(z)=z^{(N-M)}\frac{b_0z^M+b_1z^{M-1}+\cdots+b_M}{a_0z^N+a_1z^{N-1}+\cdots+a_N}$$

➢The representation form of system functions

(3) Zero point, pole, and gain constant representation

$$H(z) = Kz^{(N-M)} \frac{(z-z_1)(z-z_2)\cdots(z-z_M)}{(z-p_1)(z-p_2)\cdots(z-p_N)}$$

(4) Second order factor representation

$$H(z) = \prod_{i=1}^{L} \frac{b_{0i} + b_{1i}z^{-1} + b_{2i}z^{-2}}{a_{0i} + a_{1i}z^{-1} + a_{2i}z^{-2}}$$

System functions and unit impulse response



$$H(z)=\frac{Y(z)}{X(z)}=\frac{\mathcal{Z}\{h[k]\}}{\mathcal{Z}\{\delta[k]\}}=\mathcal{Z}\{h[k]\}$$

$$h[k]\overset{\mathcal{Z}}{\longleftrightarrow}H(z)$$

Time Domain Description of Discrete LTI Systems

Z-domain description of discrete LTI systems

$$H(z)=\frac{b_0+b_1z^{-1}+\cdots+b_{M-1}z^{-(M-1)}+b_Mz^{-M}}{a_0+a_1z^{-1}+\cdots+a_{N-1}z^{-(N-1)}+a_Nz^{-N}}=\mathcal{Z}\{h[k]\}$$

When $N=0$, $a_0\neq0(a_0=1)$,

$$H(z)=b_0+b_1z^{-1}+\cdots+b_{M-1}z^{-(M-1)}+b_Mz^{-M}=\sum_{n=0}^{M}b_nz^{-n}$$

$$h[k]=b_0\delta[k]+b_1\delta[k-1]+\cdots+b_M\delta[k-M]$$

Limited length

This discrete system is called FIR (Finite Impulse Response) system

$$H(z)=\frac{b_0+b_1z^{-1}+\cdots+b_{M-1}z^{-(M-1)}+b_Mz^{-M}}{a_0+a_1z^{-1}+\cdots+a_{N-1}z^{-(N-1)}+a_Nz^{-N}}=\mathcal{Z}\{h[k]\}$$

$N>0$， $\{a_0\neq0$, and a_k ;$k=1,2,\cdots,N\}$ has at least one non-zero term,,

$$H(z)=K\frac{(z-z_1)(z-z_2)\cdots(z-z_M)}{(z-p_1)(z-p_2)\cdots(z-p_N)}=\sum_{i=1}^{n}\frac{A_iz}{z-p_i}$$

$$h[k]=\mathcal{Z}^{-1}\{H(z)\}=\sum_{i=1}^{N}A_i(p_i)^ku[k]$$

Unlimited length

This discrete system is called an IIR (Infinite Impulse Response) system

Time domain: The necessary and sufficient conditions for the stability of a discrete LTI system are:

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

Judging the stability of a discrete LTI system by the system function $H(z)$:

If the convergence domain of the system function $H(z)$ includes the z-plane unit circle, then the system is stable.

Exception: For a causal system, if all the poles of $H(z)$ are within the unit circle, the system is stable.

E.g: Attempt to assess the stability of causal discrete LTI systems $H(z)=\frac{1}{(1-0.5z^{-1})(1-1.5z^{-1})}$

Solution: (1) Due to the system being an LTI system, analysis from the perspective of convergence domain

Because the discrete LTI system is a causal system, its convergence domain should be $|z|>1.5$

(2) Due to the fact that the system is a causal LTI system, it is analyzed from the perspective of pole distribution

The convergence domain does not contain the unit circle, so the system is unstable.

The poles of the discrete LTI system are $z_1=0.5$, $z_2=1.5$

The pole $z_2=1.5$ is outside the unit circle, so the system is unstable.

1、 The system function $H(z)$ of a stable discrete LTI system is known to be

$$H(z)=\frac{6z(z-1)}{(z-0.5)(z-2)}$$

Determine the unit impulse response $h[k]$ of the system.

2、The system function $H(z)$ of a discrete LTI system with known causality is

$$H(z)=\frac{6z(z-1)}{(z-0.5)(z-2)}$$

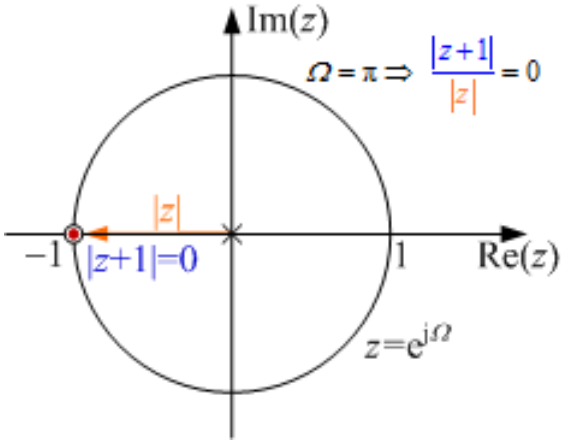
Determine the unit impulse response $h[k]$ of the system.

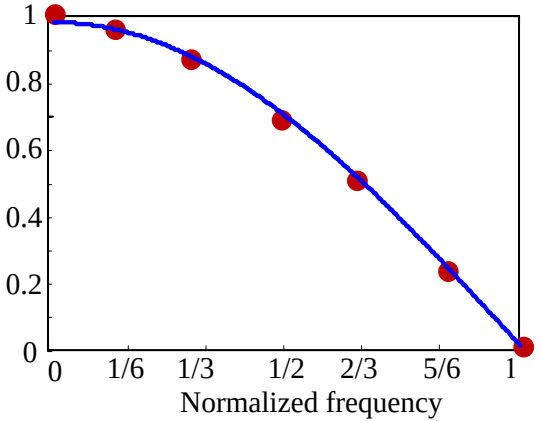
For a stable system, let $z=e^{j\Omega}$ in the system function $H(z)$ to obtain the system frequency response

$$H(e^{j\Omega}) = H(z)\Big|_{z=e^{j\Omega}}$$

E.g: Calculate the amplitude response of a first-order FIR system $H_{\mathrm{LP1}}(z) = 0.5(1+z^{-1})$

Solution: $$\left|H_{\mathrm{LP1}}(\mathrm{e}^{\mathrm{j}\Omega})\right| = 0.5\left|\frac{z+1}{z}\right|_{z=\mathrm{e}^{\mathrm{j}\Omega}}$$





Low Pass Filter

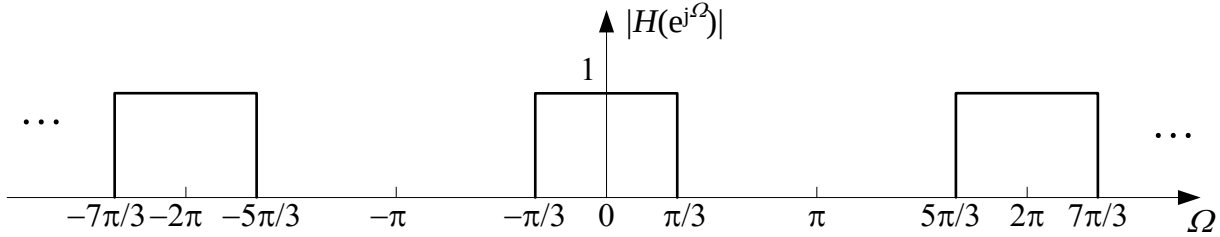
Amplitude response of first-order FIR system

Let $H(z)$ be an ideal low-pass filter with a cutoff frequency of $\Omega_c=\pi/3$, and plot the amplitude response of the following system

(1) $H_1(z)= H(-z)$ (2) $H_2(z)= H(z^2)$ (3) $H_3(z)= H(z)\,H(z^2)$

Solution:

The amplitude response of an ideal low-pass filter with a cutoff frequency of $\Omega_c=\pi/3$ is shown in the following figure



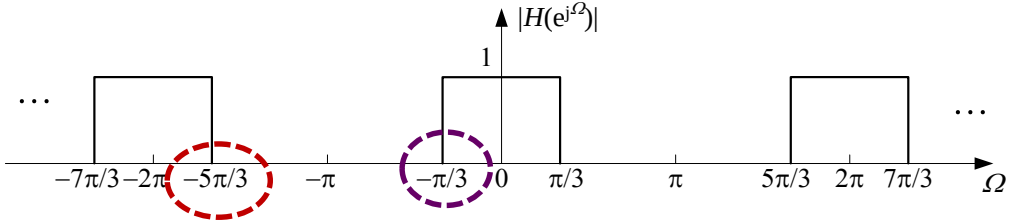
The relationship between system function and frequency response is:

$$H(e^{j\Omega}) = H(z)\Big|_{z=e^{j\Omega}}$$

Let $H(z)$ be an ideal low-pass filter with a cutoff frequency of $\Omega_c=\pi/3$, and plot the amplitude response of the following system

(1) $H_1(z)= H(-z)$ (2) $H_2(z)= H(z^2)$ (3) $H_3(z)= H(z)\,H(z^2)$

Solution:

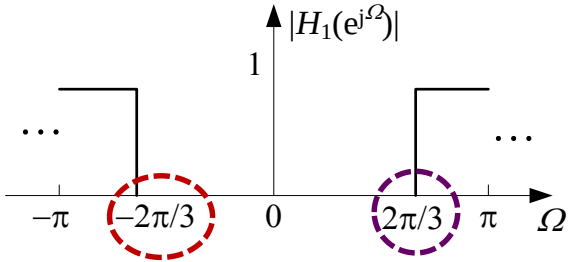


(1) $H_1(z)= H(-z)$,

$$H(e^{j\Omega}) = H(z)\Big|_{z=e^{j\Omega}}$$

$$H_1(e^{j\Omega}) = H(-e^{j\Omega}) = H(e^{-j\pi}e^{j\Omega}) = H(e^{j(\Omega-\pi)})$$

Its amplitude response is

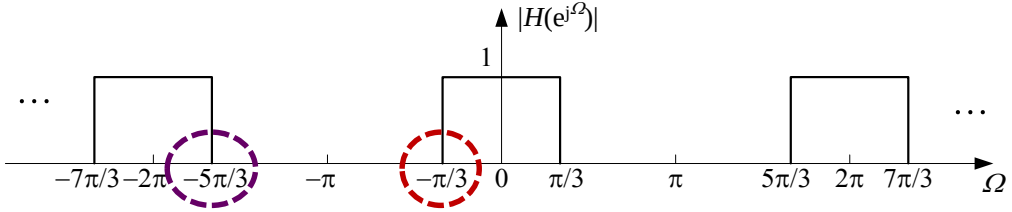


Ideal highpass filter

Let $H(z)$ be an ideal low-pass filter with a cutoff frequency of $\Omega_c=\pi/3$, and plot the amplitude response of the following system

(1) $H_1(z)= H(-z)$ (2) $H_2(z)= H(z^2)$ (3) $H_3(z)= H(z)\,H(z^2)$

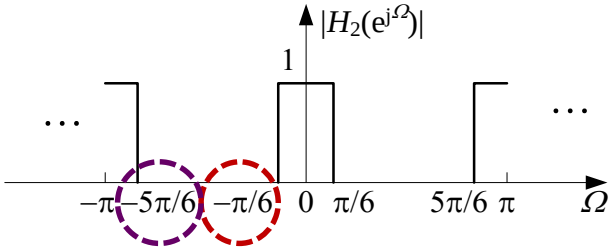
Solution:



(2) From $H_2(z)= H(z^2)$, we can obtain

$$H_2(e^{j\Omega}) = H(e^{j2\Omega})$$

Its amplitude response is

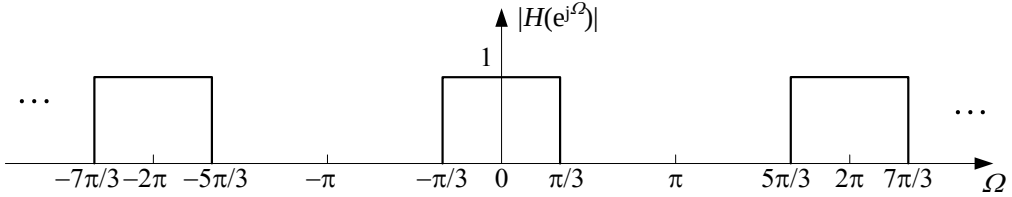


Ideal band-stop filter

Let $H(z)$ be an ideal low-pass filter with a cutoff frequency of $\Omega_c=\pi/3$, and plot the amplitude response of the following system

(1) $H_1(z)= H(-z)$ (2) $H_2(z)= H(z^2)$ (3) $H_3(z)= H(z)\,H(z^2)$

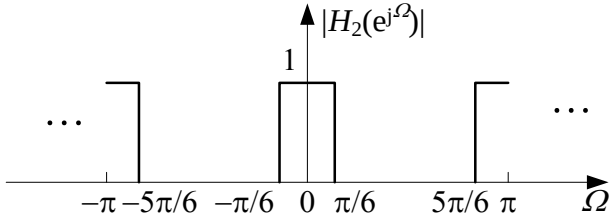
Solution:

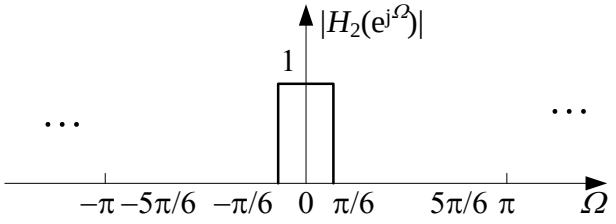


(3) From $H_3(z)= H(z)\,H(z^2)$, we can obtain

$$H_3(e^{j\Omega}) = H(e^{j\Omega})H(e^{j2\Omega})$$

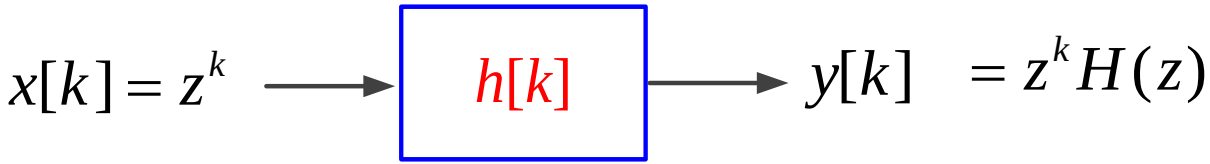
Its amplitude response is





ideal low-pass filter

System functions of discrete LTI systems:

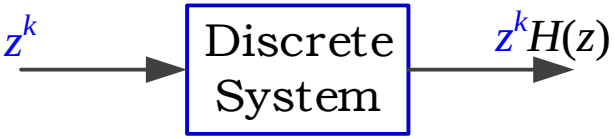


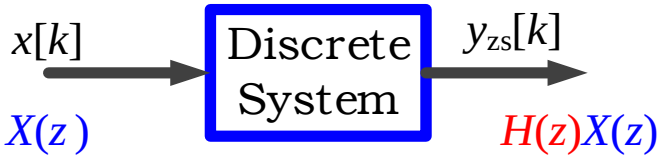
$$y[k]=z^k*h[k]=\sum_n h[n]z^{(k-n)}$$

$$=z^k\sum_n h[n]z^{-n}$$

$$=z^kH(z)$$

$H(z)$: **System functions**





$$x[k] = \frac{1}{2\pi \mathrm{j}} \oint_C X(z) z^{k-1} \mathrm{d}z$$

$$y_{zs}[k] = \frac{1}{2\pi \mathrm{j}} \oint_C X(z) H(z) z^{k-1} \mathrm{d}z$$

$$Y_{zs}(z) = H(z) X(z)$$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.